Algebraic Topology: a beginner's course

This lecture:

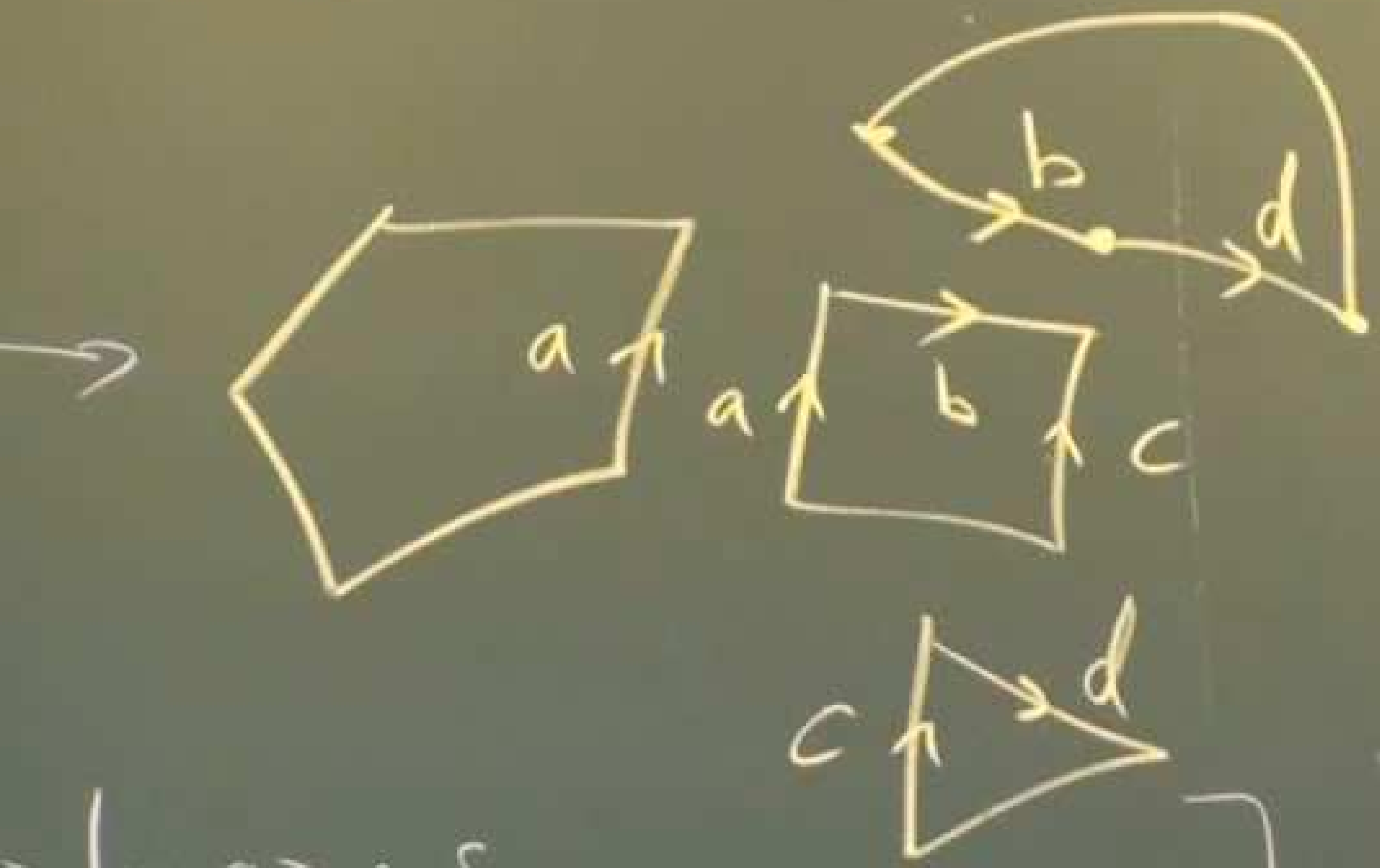
AlgTop17: Classification of combinatorial surfaces (I)


Lecture 17: Classification of combinatorial surfaces

Lecture 17. Classification of combinatorial surfaces

How to classify all

p



olygons

(2, 3, 4, ...)

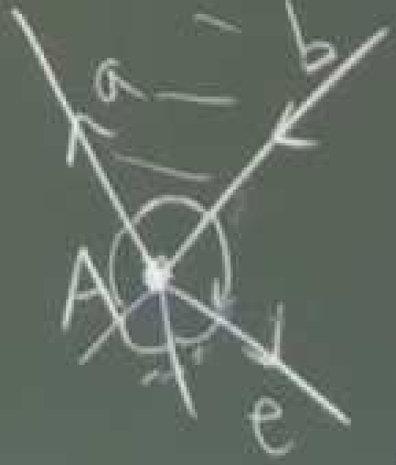
d by letters, a directed

occurs exactly twice

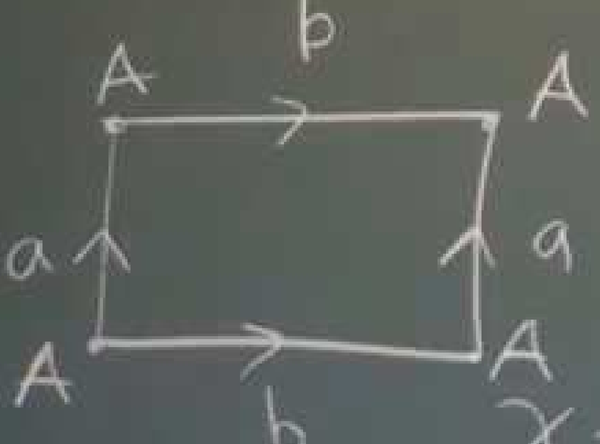
locally these n-gons fit to form a 2-dim surface:

a combinatorial surface

At a vertex get cyclic pattern of wedges ≈ open disk neighborhood

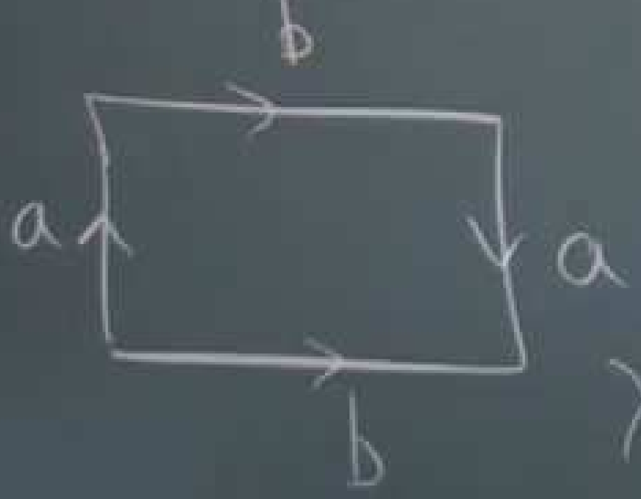


Ex. M



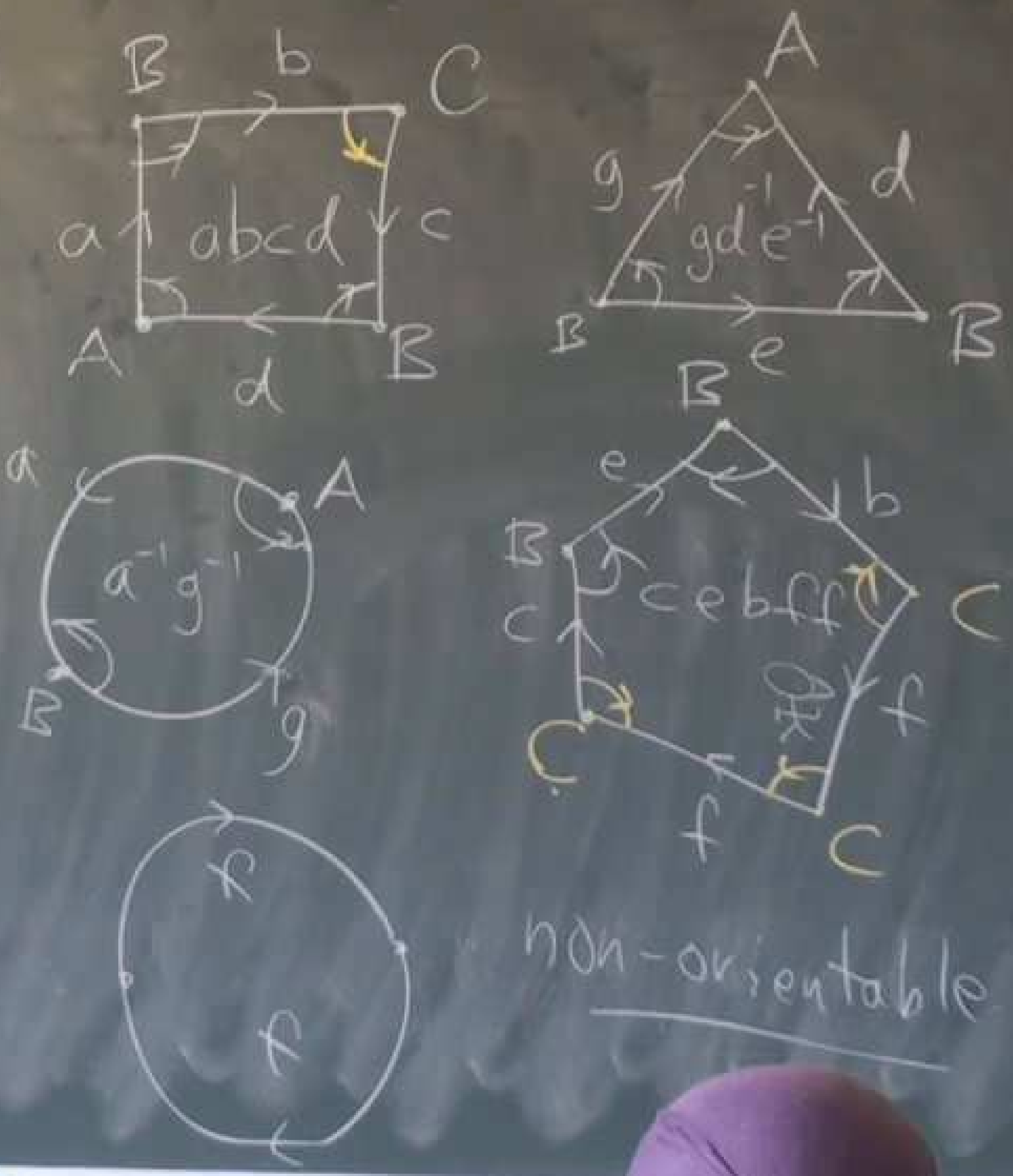
torus

$\chi = 1-2+1$
$= 0$



Klein bottle

$\chi = 1-2+1 = 0$



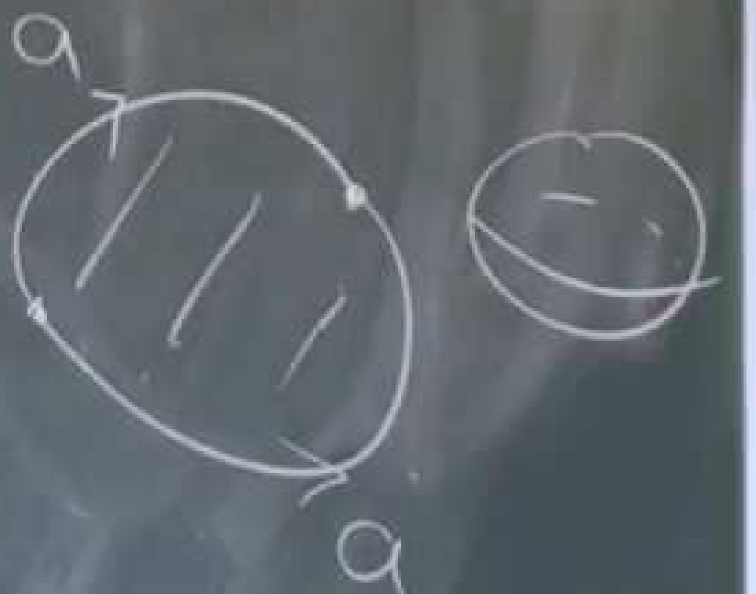
non-orientable

Euler n
of M
$= \chi = V -$
$= 3 - 7$
$= 0.$

Thm (Classification of connected combinatorial surfaces)

Every connected combinatorial surface can be represented in exactly one of the following standard forms:

1. [Sphere] $M = a a^{-1}$



2. [Sum of tori / n-holed torus]

$$M = a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1} \cdots a_n b_n a_n^{-1} b_n^{-1}$$

...binatorial

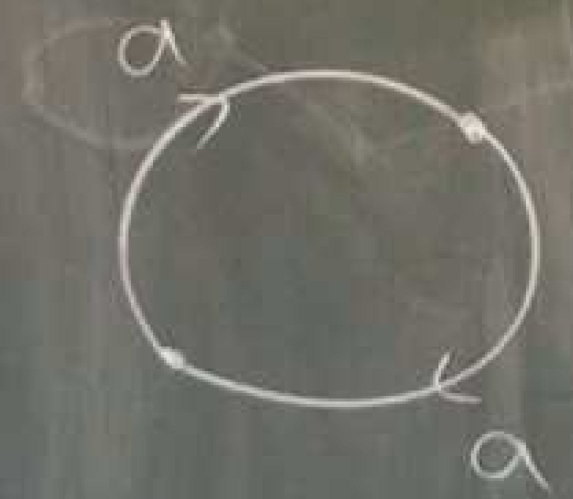
...face

...e of the

3. [Sum of k projective planes/crosscaps]

$$\mathcal{M} = a_1 a_1 a_2 a_2 \cdots a_k a_k$$

$$\chi = 2 - k$$



$aa \equiv$ proj plane

Moreover, $\mathcal{M}$ is then determined by its Euler number together with orienta...

rus

$-2+1$

0

lein!

$-2+1$

Idea of proof:

torus

$= 1 - 2 + 1$
$= 0$

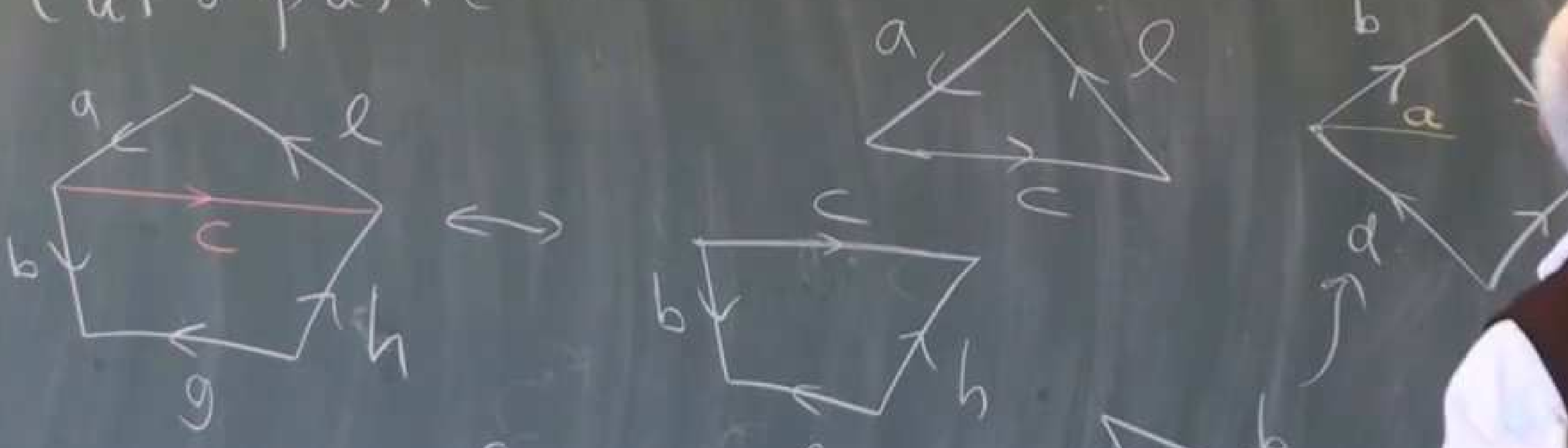
Klein

$\chi = 1 - 2 + ($

Idea of proof: Series of reductions to a normal form:

$bkf^{-1}d$

'Cut & paste':

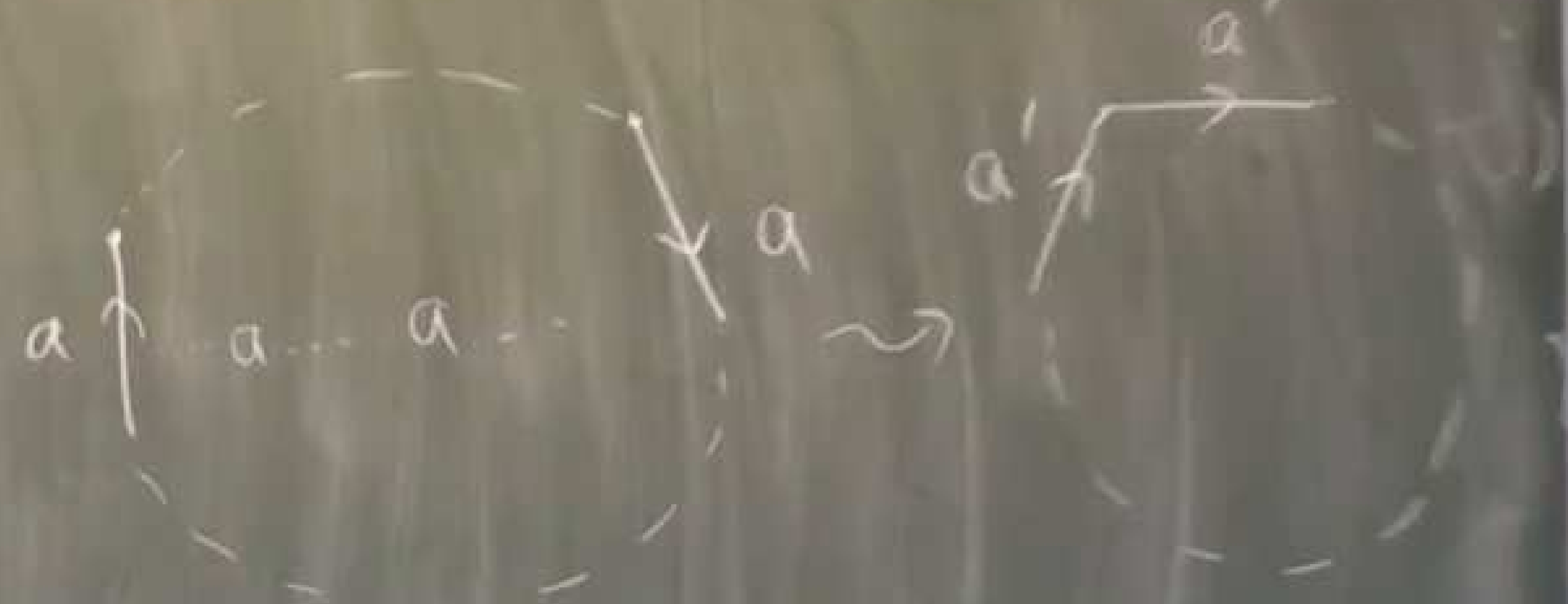


Reduce:

gons together to

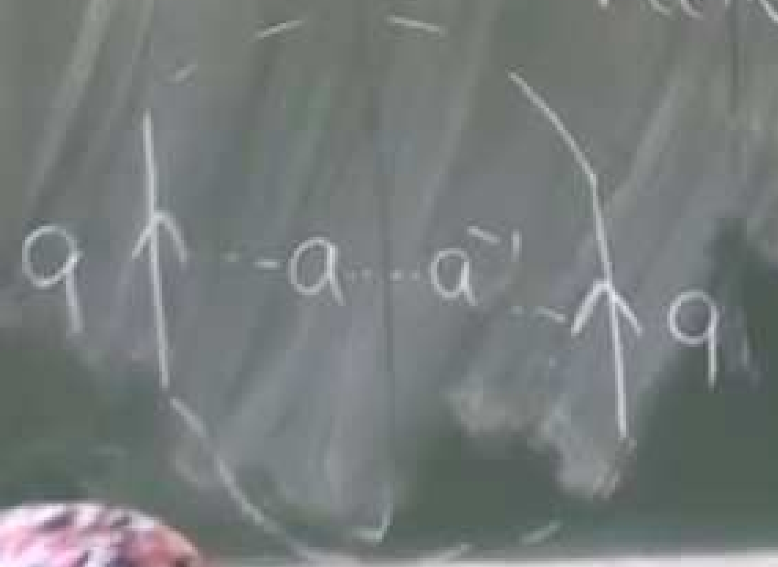
gon

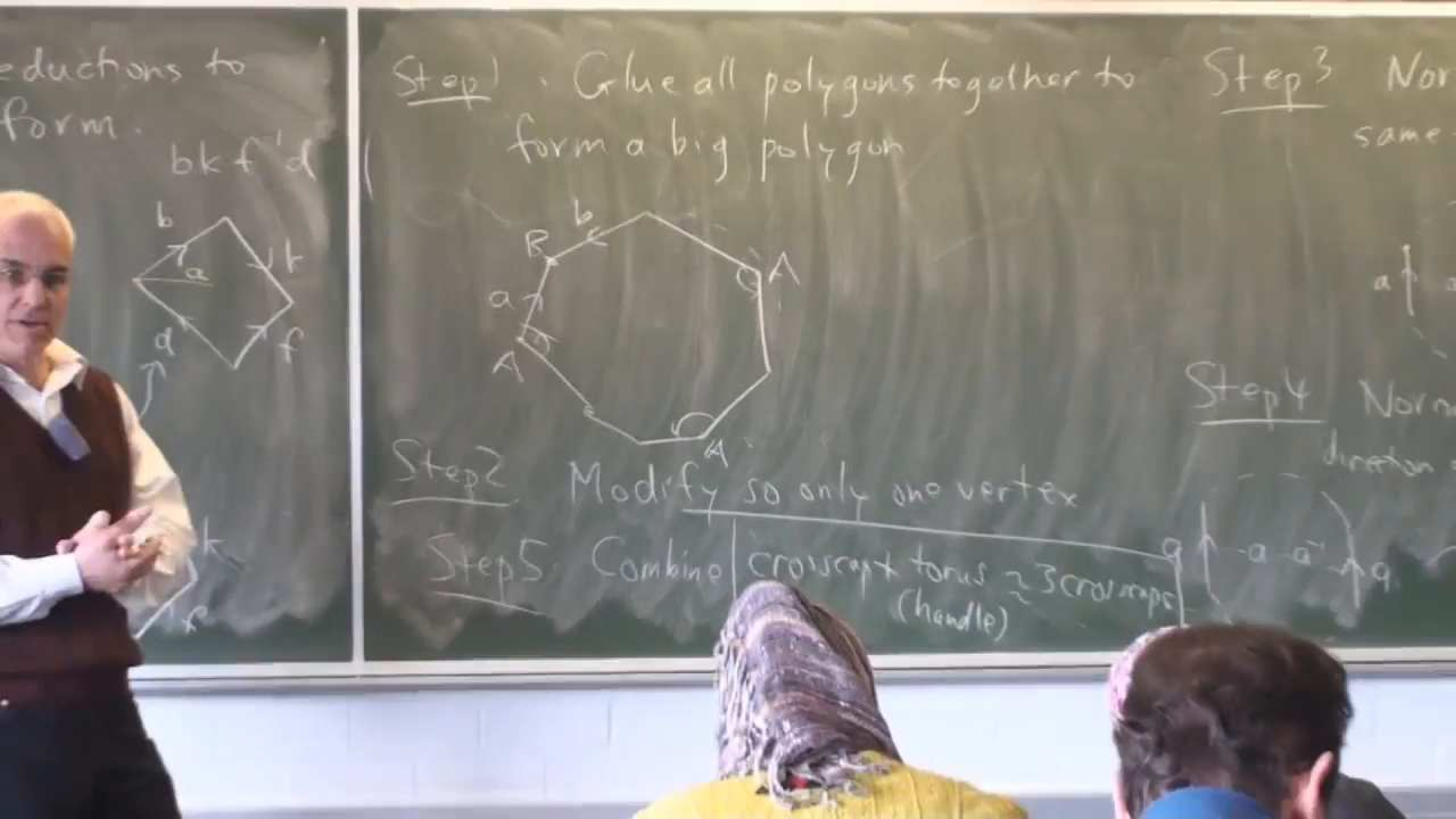
Step 3 Normal form for same-direction edges.

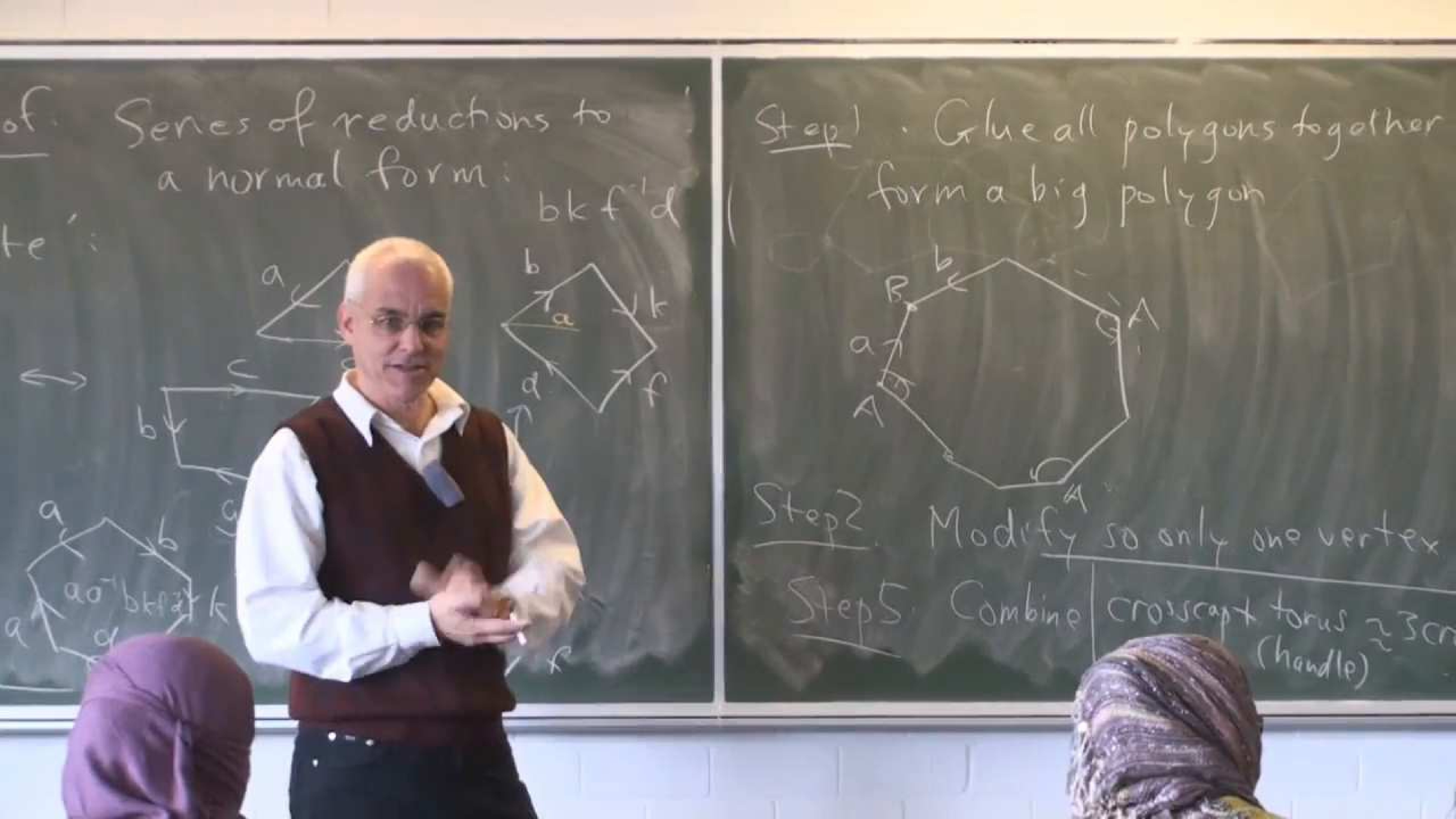


Step 4 Normal form for opposite direction edges

one vertex




eductions to
form.
Step 1 . Glue all polygons together to
form a big polygon
Step 2 Modify so only one vertex
Step 3
Step 4
Step 5 Combine
crosscap + torus ≅ 3 crosscaps
(handle)


of. Series of reductions to
a normal form:
bkf⁻¹d
te':
aa⁻¹bkf⁻¹d⁻¹k
Step1. Glue all polygons together
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crosscap + torus ≃ 3cr
(handle)